



# Datasheet

## RGB-621-R1 RGB Lighting Control Module

Part No.: RGB-621-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + Field Board)

Controller: RP2350A



## Functional Overview

The RGB-621-R1 is a DIN-rail mounted RGBCCT lighting control module for 12-24 V DC LED systems. It provides five low-side MOSFET LED outputs (R, G, B, CW, WW), two 24 V digital inputs, one relay output, RS-485 Modbus RTU communication, and USB-C configuration interface.

| Board       | Function                                                                    |
|-------------|-----------------------------------------------------------------------------|
| MCU Board   | RP2350A controller, RS-485 communication, USB-C interface, power conversion |
| Field Board | MOSFET LED outputs, Relays, DI                                              |

The module operates as a Modbus RTU slave over RS-485 and supports configuration and firmware update via USB-C.

## Technical Specifications

| Specification                             | Details                                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Microcontroller                           | RP2350A dual-core microcontroller                                                                                                                             |
| Storage                                   | External QSPI Flash (32 Mbit)                                                                                                                                 |
| Power Input                               | 24 V DC nominal (18–30 V DC recommended)                                                                                                                      |
| Input Protection                          | 1 A time-lag fuse, reverse polarity diode, TVS surge suppression, EMI filtering                                                                               |
| Main Logic Supply                         | Buck regulator 24 V → 5 V (~4.96 V nominal), 3.3 V LDO regulator                                                                                              |
| LED Outputs                               | 5 × low-side MOSFET outputs (R, G, B, CW, WW)                                                                                                                 |
| LED Output Voltage                        | External 12-24 V DC supply                                                                                                                                    |
| Maximum Total LED Current                 | Limited by 5 A input fuse (shared across channels)                                                                                                            |
| Recommended Current per Channel           | ≤ 1000 mA per channel (recommended)                                                                                                                           |
| Relay Outputs*                            | 3 × SPDT dry-contact relays                                                                                                                                   |
| Rated Load (System Limit)                 | 3 A @ 250 VAC (resistive contact rating)                                                                                                                      |
| Max Load @ 250 VAC                        | 750 VA (system limit)                                                                                                                                         |
| Max Load @ 30 VDC                         | 90 W (system limit)                                                                                                                                           |
| Relay Component Rating (Informative Only) | Relay components are rated up to 12 A @ 250 VAC (resistive); however, this rating does not apply to the complete module                                       |
| Digital Inputs                            | 2 × isolated 24 V DC discrete inputs (DIx + GND), supporting dry-contact closure protected by per-channel PTC fuse, TVS surge suppression, and EMI filtering. |
| User Interface                            | 2 buttons, 8xLEDs(Power, 2 user LEDs, RX, TX, 1xRelay, 2xDI)                                                                                                  |
| RS-485                                    | Half-duplex Modbus RTU with surge protection and fail-safe biasing                                                                                            |
| USB                                       | USB-C, 5 V logic, ESD protected                                                                                                                               |
| Typical Power Consumption                 | 0.2–0.5 W                                                                                                                                                     |

\*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

Typical module power consumption: 0.2–0.5 W (logic only).

Maximum module power consumption: approximately 1W.

## Top Terminal Block (1–9)

Front view – terminals numbered from left to right.

| Pos | Marking | Signal | Description              |
|-----|---------|--------|--------------------------|
| 1   | +       | LED_V+ | 24 V DC LED supply input |

|   |     |           |                             |
|---|-----|-----------|-----------------------------|
| 2 | -   | LED_0V    | LED power return            |
| 3 | V+  | +24V      | 24 V DC module supply input |
| 4 | 0V  | 0V        | Module power return         |
| 5 | NO  | RELAY_NO  | Relay normally open         |
| 6 | C   | RELAY_COM | Relay common                |
| 7 | I1  | DI1       | Digital input 1             |
| 8 | GND | DI1_GND   | Digital input return        |
| 9 | I2  | DI2       | Digital input 2             |

## Bottom Terminal Block (9–18)

Front view – terminals numbered from left to right.

| Pos | Marking | Signal    | Description                              |
|-----|---------|-----------|------------------------------------------|
| 10  | R       | LED_R     | Red LED channel (low-side output)        |
| 11  | G       | LED_G     | Green LED channel (low-side output)      |
| 12  | B       | LED_B     | Blue LED channel (low-side output)       |
| 13  | CW      | LED_CW    | Cool white LED channel (low-side output) |
| 14  | WW      | LED_WW    | Warm white LED channel (low-side output) |
| 15  | COM     | LED_COM   | LED common positive (+24 V)              |
| 16  | A       | RS485_A   | RS-485 A line                            |
| 17  | B       | RS485_B   | RS-485 B line                            |
| 18  | COM     | RS485_COM | RS-485 COM                               |

## Installation, Environmental & Mechanical

| Specification      | Details                                        |
|--------------------|------------------------------------------------|
| Terminal Type      | Pluggable screw terminal blocks, 5.08 mm pitch |
| Wire Cross Section | 0.2–2.5 mm <sup>2</sup> (AWG 24–12)            |
| Tightening Torque  | 0.4–0.6 Nm                                     |

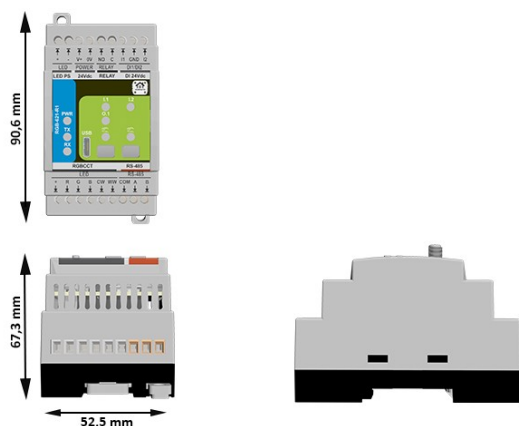
| Specification         | Details                    |
|-----------------------|----------------------------|
| Operating Temperature | 0 °C to +40 °C             |
| Storage Temperature   | –10 °C to +55 °C           |
| Relative Humidity     | 0–90 % RH, non-condensing  |
| Ingress Protection    | IP20 (inside cabinet only) |
| Maximum Altitude      | 2000 m                     |
| Pollution Degree      | 2                          |



| Specification      | Details                                                     |
|--------------------|-------------------------------------------------------------|
| Dimensions         | 52.5 × 90 × 59 mm (L × W × H)                               |
| DIN units          | 3 division units ( ≈ 53 mm DIN rail mounting width)         |
| Net weight         | Xxx g                                                       |
| Gross weight       | Xxx+98 g                                                    |
| Pack size          | 140× 125 × 94 mm (L × W × H)                                |
| Mechanical drawing | Front + side views with DIN-clip depth (see below)          |
| Notes              | All dimensions shown in millimeters unless stated otherwise |
| Mounting           | 35 mm DIN rail (EN 50022)                                   |
| Enclosure Material | PC/ABS, UL94-V0                                             |

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.



## Cable Recommendations & Shield Grounding

This section applies to RS-485. Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

## General Routing Rules

- Route low-level signal cables separately from power wiring.
- If crossing power cables is unavoidable, cross at 90°.
- Keep cable runs as short as practical.
- Avoid parallel runs with high-current conductors.

## RS-485 Cable

- Twisted pair recommended (120  $\Omega$  characteristic impedance).
- Shielded cable recommended in industrial environments.
- Use one twisted pair for A/B.
- Use second conductor as COM reference if required.

Examples: J-Y(ST)Y 2×2×0.5 mm<sup>2</sup> or LI2YCY PiMF 2×2×0.50

## Shield Grounding

- Bond cable shield(s) to cabinet PE/EMC ground at one end (PLC side recommended).
- Do not connect shields directly to signal terminals (A/B/COM).

## Compliance & Certifications

The RGB-620-R1 module is CE marked and designed to comply with applicable European Union directives. The manufacturer maintains technical documentation and a signed EU Declaration of Conformity (DoC).

EMC 2014/30/EU · LVD 2014/35/EU · RoHS 2011/65/EU

Standards: EN 61000-6-1 · EN 61000-6-3 · EN 62368-1 · EN IEC 63000

## Safety Notice

The power input and signal terminals are SELV circuits.

Installation by qualified personnel only.

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